



How to Process your BESS Availability Data

How to incorporate Energy Capacity into BESS Availability

Module 1 - BESS Availability Explained

 **by Fred Rivollier**



OBE – Overall Battery Effectiveness

Like OEE in manufacturing but built for BESS

$$OBE = \frac{1}{N} \sum_{i=1}^N \min(EA_i, OA_i, PA_i, QA_i)$$

EA

Equipment Availability

OA

Operational Availability

PA

Power Availability

QA

Energy Availability

OBE is the average, across all intervals, of the minimum availability recorded in each interval

The weakest of the four types sets the ceiling at every point in time.

The Availability Data Journey from EMS to your Inbox



example - Performance Engineer AI Agent

explore data | build Grafana Dashboards | Build Marimo Notebooks

claude_performance_engineer / Document available skills and capabilities

/perf-engineer

Read performance-engineer.md

BESS Performance Engineer mode active.

I'm ready to diagnose BESS asset performance using the Synapse ClickHouse platform. I have the full cookbook loaded (PCS faults, SOC imbalance, thermal, availability, degradation, efficiency) and direct access to the data via mcp__mcp-clickhouse__* tools.

What performance concern are you investigating today?

1. PCS faults / status — fault code decode, status transitions, Pareto

2. SOC imbalance — string SOC spread, cell voltage dispersion, BMS health

3. Thermal issues — hot strings, derating, HVAC effectiveness

4. Availability / reliability — gap-detection method, PCS/string availability

5. Degradation — SOH trending, capacity fade, resistance growth

6. Power / efficiency — AC vs DC power, round-trip efficiency, parasitic losses

7. Cross-site benchmarking — normalized fleet comparisons

8. Something else — describe the symptom and I'll pick the right query path

Also let me know:

Type / for commands

Accept edits + - Opus 4.7 1M · Extra high

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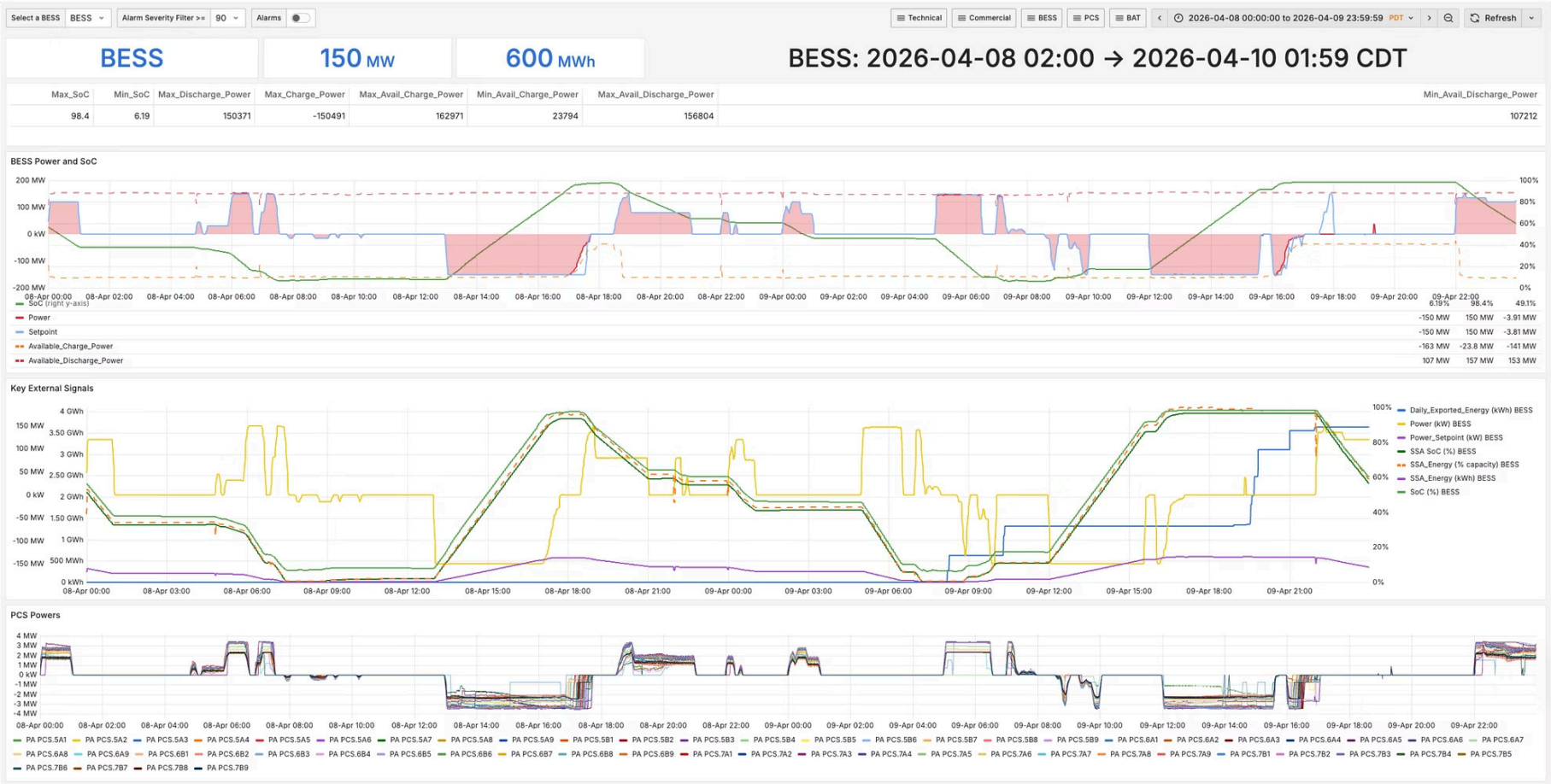
Top PCS by Fault Count

Rank	PCS	Fault Count	Unique Codes	Codes Seen
1	PCS.7B2	6	4	89, 150, 140, 350
2	PCS.6A3	5	3	0, 57, 45
3	PCS.6A6	5	3	0, 13, 350
4	PCS.6A5	5	3	350, 0, 17
5	PCS.6A7	4	2	0, 150
6	PCS.4A3	4	1	57 (repeat)
7	PCS.2A2	3	3	13, 35, 150
8	PCS.3A8	3	3	191, 350, 13
9	PCS.6A4	3	2	45, 0
10	PCS.5B5	2	2	350, 13

Type / for commands

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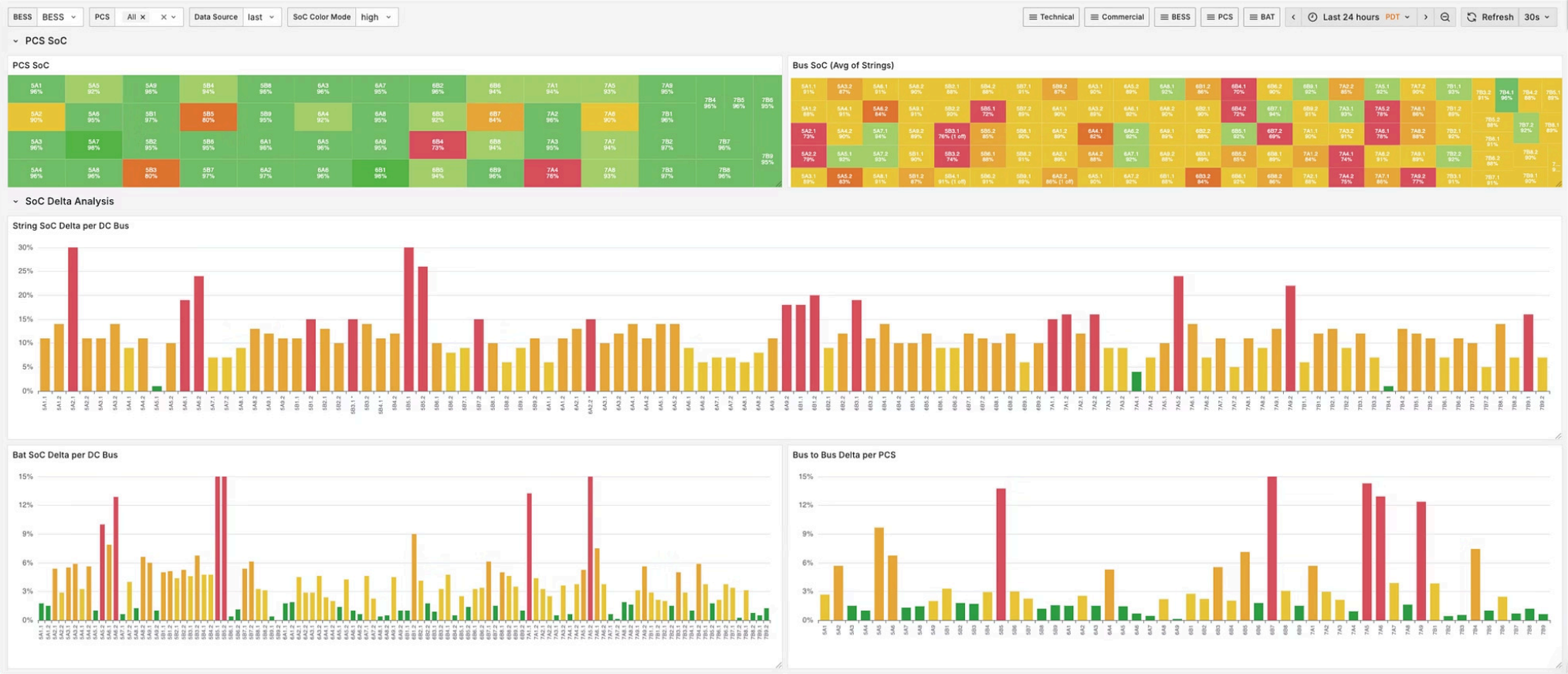
example of operational dashboard



example availability dashboard



example balancing monitoring dashboard



Marimo Notebook Demo

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The future of Python notebooks **is here**

Transform data, train models, and run SQL queries with marimo — feels like an AI-native reactive notebook, stored as Git-friendly reproducible Python. Seamlessly run as scripts and apps. All open source.

```
$ pip install marimo  
$ marimo tutorial intro
```



That's a Wrap on Module 1

Module 1 covered BESS Availability end-to-end — the full OBE framework and how to actually compute and report it.

- Equipment Availability — what's physically online
- Operational Availability — are we meeting setpoint
- Power Availability — can we deliver full power when called
- Energy Availability — do we still have the capacity we were promised
- Automating the reporting — from EMS data to daily report

You now have the vocabulary, the math, and the processing pattern to stand up availability reporting on your own fleet.



Module 2 - Performance Engineering as a Discipline

Availability tells you what happened. Performance engineering is how you run the practice that catches it, explains it, and closes it out.

Module 2 zooms out from availability and into the discipline itself:

What performance engineering actually is and what it isn't

The three pillars: data infrastructure, daily engineering discipline, investigation management

The process: how the work flows from telemetry to action

What it takes to build and run this practice on a real BESS fleet



Thank you for joining me on this exploration of BESS Availability and the foundations of Performance Engineering.

Stay tuned for future modules and hands-on demonstrations!

